

DIABETIC RETINOPATHY DETECTION USING SUPERVISED LEARNING

20BD1A6704 ATHARVA RAJE

20BD1A6706 ROHIT BANDARI

20BD1A6707 B SAKETH REDDY

20BD1A6721 GANUMALA L LALIT

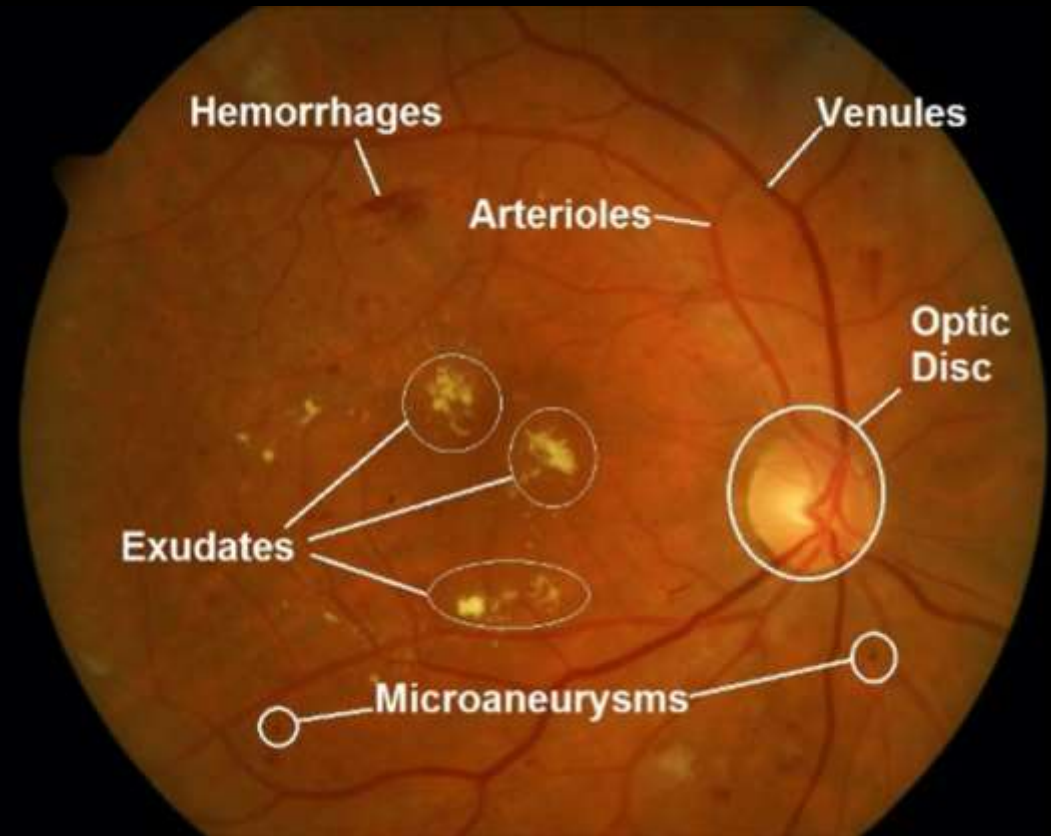


ABSTRACT- THE CRUX

Diabetic Retinopathy is one of the leading causes of blindness and eye disease in the working-age population of the developed world. This project attempts to find an automated way to detect this disease in its early phase. In this project, I am using supervised learning methods to classify a given set of images into 5 classes. For this task, I am employing various image processing techniques and filters to enhance many important features and then using neural for classification. Diabetic retinopathy causes blindness. Diabetic retinopathy is one of the eye diseases which is caused due to retinal blood vessel extraction. Diabetic retinopathy affects blood vessels in the light-sensitive tissue called the retina that lines the back of the eye. It is the most common cause of vision loss among people with diabetes and the leading cause of vision impairment and blindness among working-age adults. Here we proposed a system where we extract retinal blood vessels for detecting eye diseases. Manually extracting the retinal blood vessels is a long task. There are many automated methods available that make work easier. Here proposed an algorithm to extract blood vessels from images. We used filtering methods to remove noise and environmental interference from the image. Local entropy thresholding and alternative sequential filter methodology have been adopted in this system. We had implemented this system in Matlab. The user will input a retina image into the system. The system will apply filtering techniques. Image pre-processing steps are applied to get accurate results. System will remove all unwanted objects from the image. System will apply an algorithm to extract retinal blood vessels. Finally the system will detect diabetic retinopathy.

INTRODUCTION

Diabetic retinopathy is a case seen in diabetes induced patients where the patient suffers loss of vision due to block blood vessels.

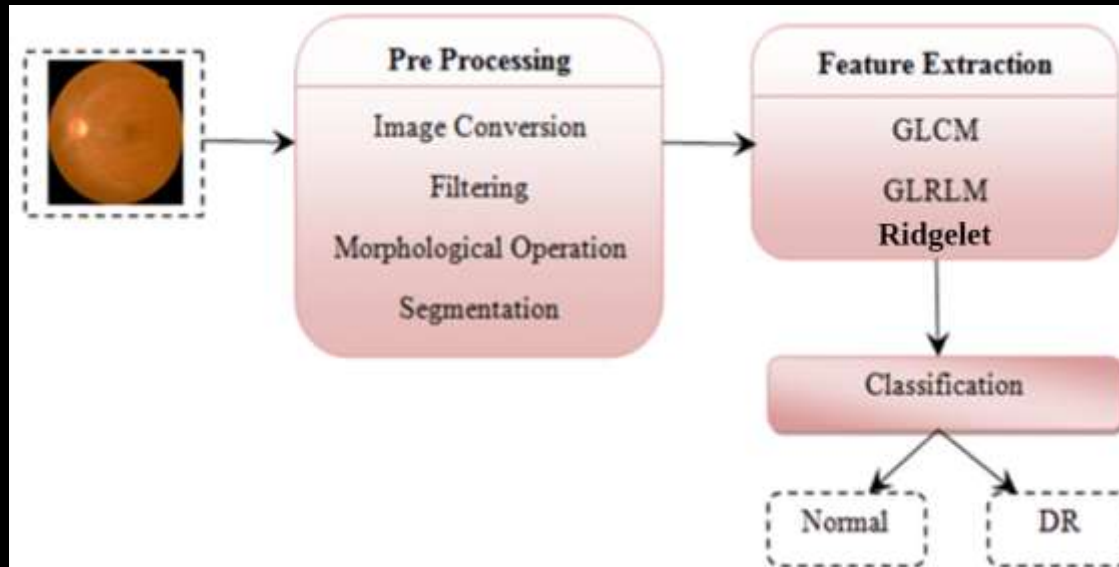


Traditional way of detecting diabetic retinopathy

In traditional way of detecting diabetic retinopathy, the ophthalmologists use manual machines to study the eye in several angles which is a tedious and difficult procedure. Its reliability also is highly questionable

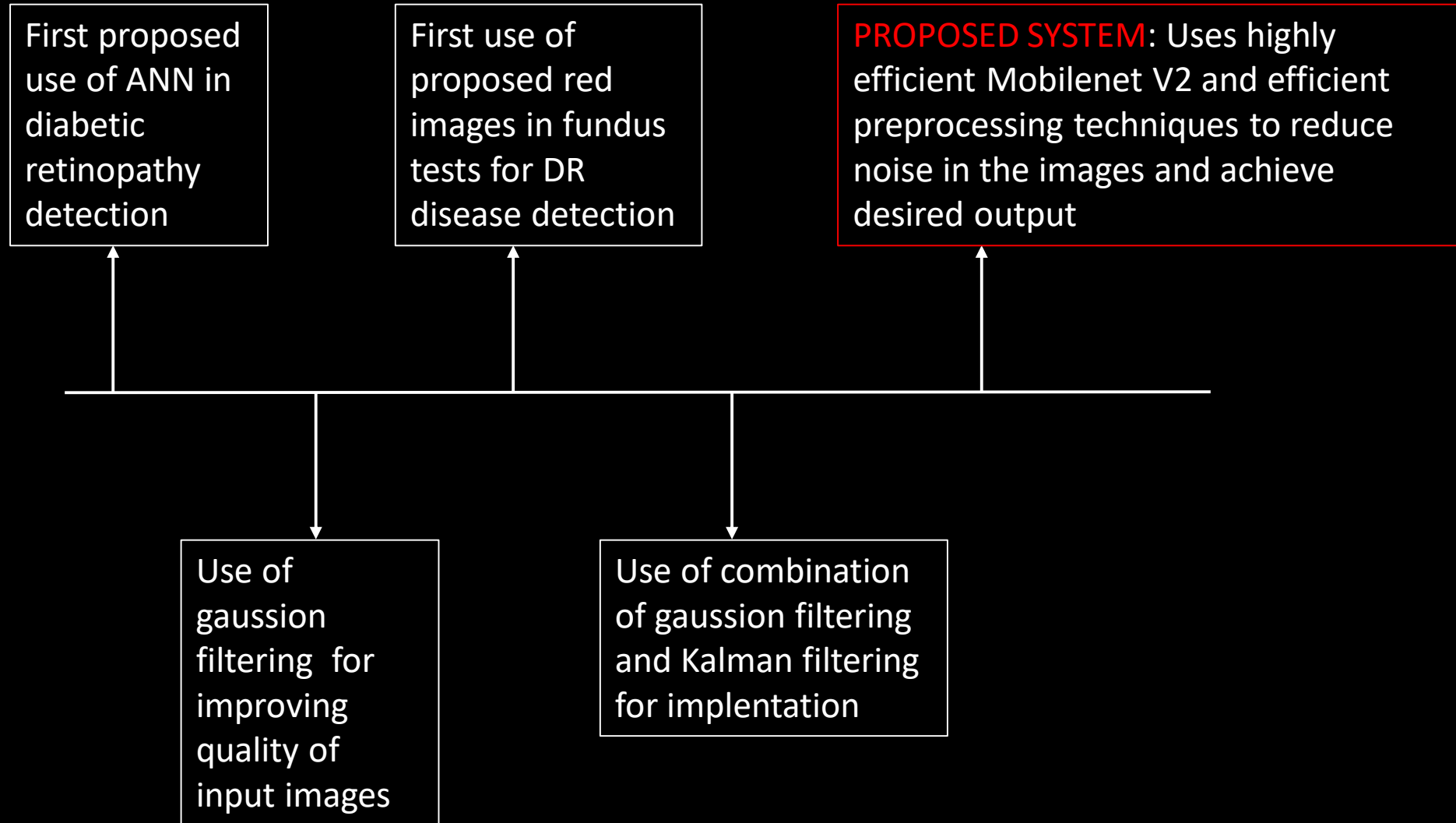


Proposed way of doing solving the problem



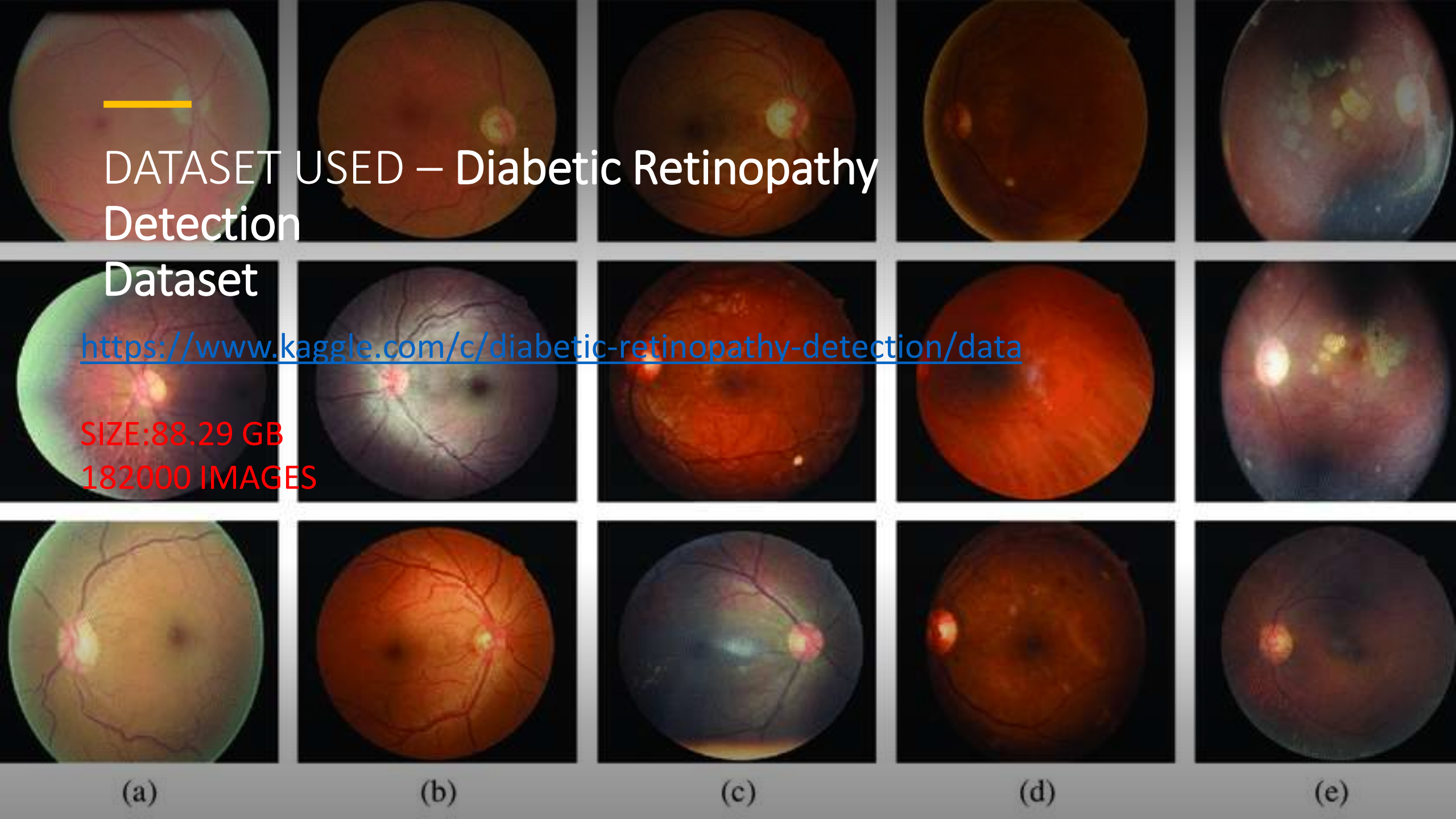
The proposed way is fully automated. This involves using deep neural networks to analyze the images and detect blocks in the blood vessels. Usage of Kalman filtering to statistically reduce noise in images. We also propose to use MobileNetV2 and bigger dataset.

Literature survey – A timeline of trials



History of Records

YEAR	PROPOSED WAY	DATASET and ACCURACY
1999	Applied a deep learning to learn an algorithm for automatically detection of DR.	EyePACS-1 included 963 images from 4997 patients ACCURACY- 76%
2008	DCNN for detecting microasyrums were used	Modified custom dataset collected from random patients ACCURACY-73%
2016	Use of pretrained CNN model. Primarily VGG16 and VGG19.	ACCURACY (VGG16) - 71.7 (VGG19) - 73.9%
Present*	Use of MobileNetV2.	Diabetic Retinopathy Dataset- (182000 images) Estimated accuracy- 92%



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DATASET USED – Diabetic Retinopathy Detection Dataset

<https://www.kaggle.com/c/diabetic-retinopathy-detection/data>

SIZE:88.29 GB
182000 IMAGES

(a)

(b)

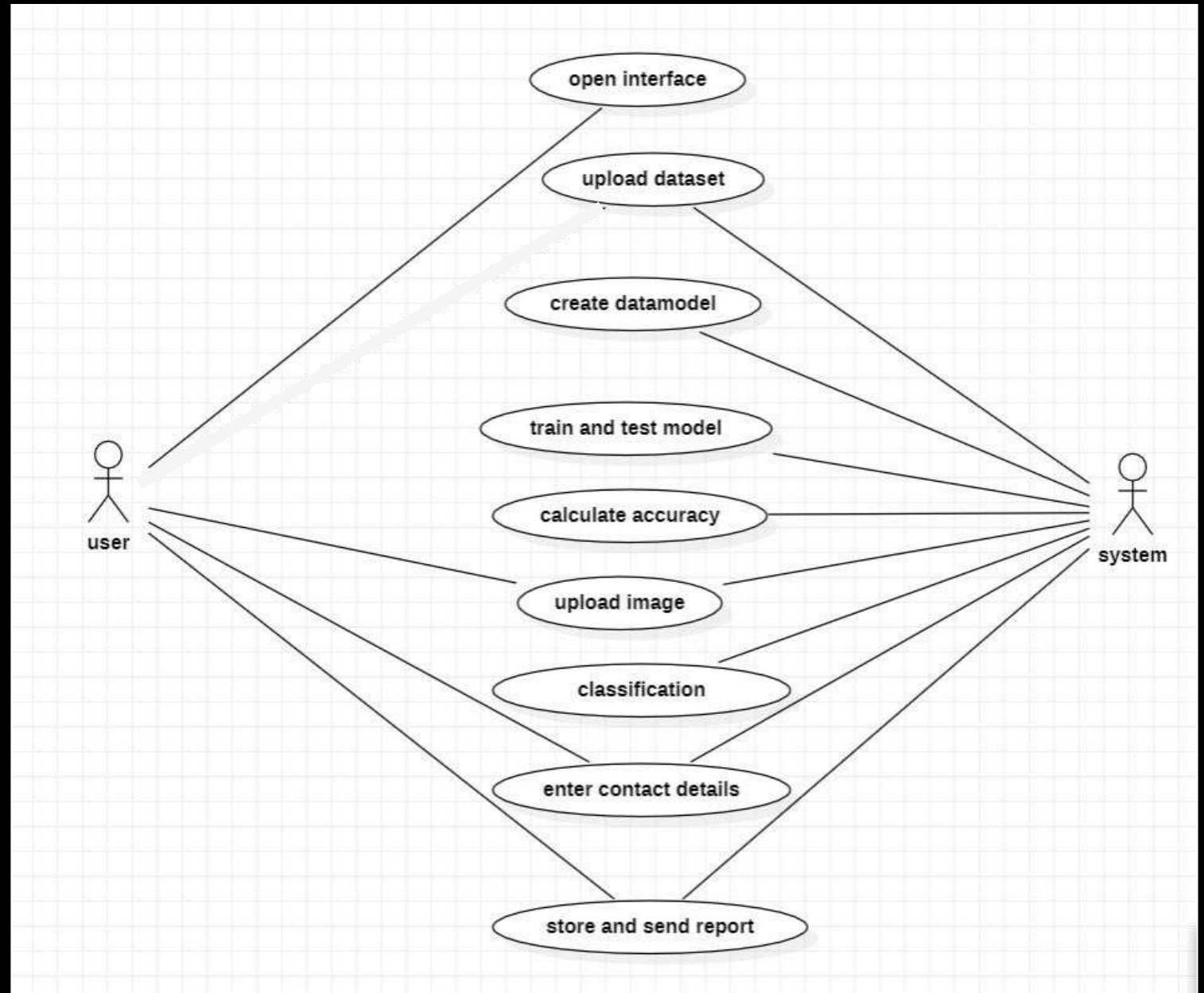
(c)

(d)

(e)

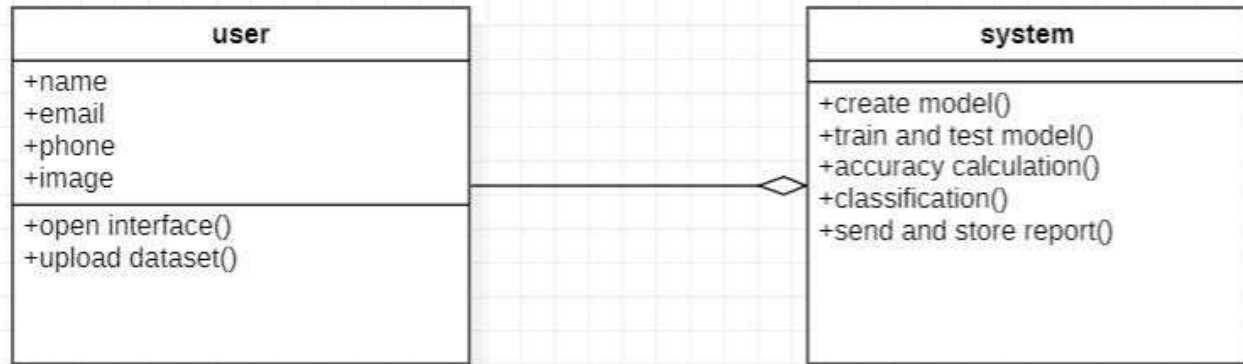
Use case diagram:

Design of the project – UML Architecture



Class diagram:

Design of the project – UML Architecture



THANK YOU

